Geometry-MMALS G1 v1.1.2

Smooth Simplex-Residual Routing and Covariance-Aware Context Diagnostics

Five-seed results, corrected manifests, falsification outcome, and reviewer-facing interpretation

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RotatedMNIST pilot, five matched-compute seeds, June 2026

Epistemic status. The five-seed pilot completed for seeds 0–4 under the frozen bridge-isolation protocol. The primary R5 smooth-residual router did not qualify against the preregistered R1 linear structural reference: held-out functional order, held-out functional stress, continuity non-degradation, causal specificity, and operational non-degradation did not all pass. However, R5 is consistently more ordered and lower-stress than the flexible R0 MLP router, route-swap ordering passes, and the sharp local tails of R3 are removed. The covariance-aware SPD diagnostic also passes: the combined centroid-plus-Log-Euclidean context metric reduces stress relative to centroid-only geometry in all five seeds. This SPD result is diagnostic and does not establish a learned Riemannian manifold.

Abstract

Geometry-MMALS asks whether an inferred context geometry can organize routing through a modular host ecosystem in a functionally meaningful and auditable way. Version 1.1.2 preserves the strict bridge-isolation controls of v1.1.1: within each seed, the sensory encoder, four-dimensional context encoder, host bank, synthesis normalization, classifier, and common host-functional cost matrix are frozen and shared across all router treatments. We compare six matched-compute routers: a flexible MLP (R0), a linear router (R1), prototype energy (R2), directional-prototype hybrid energy (R3), a bounded smooth residual router (R4), and the same router with a context-only simplex-continuity hinge (R5).

The primary R5-versus-R1 held-out functional effects are small and non-qualifying. Functional distance-order correlation changes by +0.0109 with 95% seed interval $[-0.0487, 0.0705]$, and functional stress changes by -0.0179 with interval $[-0.0671, 0.0314]$. Local functional continuity q95 is nearly identical on average for R5 and R1 (0.2693 versus 0.2713), but the paired seed interval $[-0.1702, 0.1663]$ exceeds the preregistered non-degradation tolerance. Causal specificity fails because only three of five seeds have a minimum lower bootstrap bound above one, despite mean functional CSR 1.780. Operational non-degradation fails: R5 loses 0.964 percentage points of held-out accuracy and 0.894 points of final trained accuracy relative to R0, although forgetting improves by 0.169 points.

Two narrower results survive. First, R5 significantly improves over R0 on held-out common-cost functional geometry: $\Delta\rho = +0.0496$ with interval $[0.0243, 0.0748]$ and $\Delta\text{stress} = -0.0394$ with interval $[-0.0592, -0.0196]$. Second, R5 reduces the severe R3 continuity q95 by 1.676 on average without losing held-out functional geometry. The smooth repair therefore removes prototype sharpness but adds no robust value beyond the linear router. Separately, covariance-aware context diagnostics pass: centroid-plus-Log-Euclidean stress improves by -0.0442 with interval $[-0.0638, -0.0246]$ and a favorable fraction of 5/5 seeds. The result redirects the program away from additional router complexity and toward route-function memory transport and causal host-bank alignment.

Contents

1 Reviewer decision summary

Primary decision

The R5 smooth-residual router does not qualify. The failure is scientific rather than procedural: the common components are exactly frozen, compute is matched, and all five seeds completed. The primary R5-versus-R1 structural comparison does not establish a robust gain in held-out functional order, functional stress, or local continuity, and the operational equivalence gate fails against R0.

What survives

Three narrower findings are supported. R5 outperforms R0 on held-out route geometry; R5 removes the severe local sensitivity of R3; and covariance-aware context geometry adds stable factor information beyond centroid-only geometry. These findings constrain the next experiment even though they do not qualify R5 as the preferred router.

Table 1: Reviewer-facing gate outcome. A failed gate means the preregistered threshold was not met, not that the software or execution failed.

Gate	Outcome	Interpretation
Context and common host bank frozen	Pass	Exact zero context and host-output deltas across treatments.
Equal router compute	Pass	15,360 forward images and 80 optimizer steps per method and seed.
Held-out functional stress reduction vs R1	Fail	Mean favorable, but seed interval crosses zero.
Held-out functional order vs R1	Fail	Mean favorable, but seed interval crosses zero.
Local continuity non-degradation vs R1	Fail	Mean nearly equal, but the paired upper interval exceeds tolerance.
Route-swap order	Pass	Far swaps exceed near swaps for every R5 seed.
Functional causal specificity	Fail	Only three of five seeds have lower CSR bound above one.
Prediction identity preservation	Pass	Minimum per-seed preservation is 0.981.
Operational non-degradation vs R0	Fail	Too few seeds remain within the one-point held-out margin.
SPD covariance adds value	Pass, diagnostic	Combined covariance-aware stress improves in all five seeds.
Mature host specialization	Not tested	Hosts are frozen; only allocation is observed.
Memory transport	Not implemented	Deferred to v1.1.3.
Operational superiority	Not claimed	Not preregistered as a primary claim.

1.1 Manifest correction

The numerical run completed, but the original `claim_manifest.json` retained the pre-execution string `pilot_pending`. The release corrects this to `five_seed_pilot_completed`, records `r5_smooth_residual_router_` and records `spd_covariance_adds_value_candidate_supported`. This is a metadata-only correction. No CSV value, checkpoint hash, cost matrix, gate, or source split changed.

2 Scientific question and controlled design

2.1 Why v1.1.2 exists

Version 1.1.1 established a clean bridge-isolation protocol but rejected the prototype-dominated R3 router. The main failure mode was not an absence of route structure. R3 and R2 reduced functional stress and produced ordered far-versus-near route swaps. The dominant problem was excessive local sensitivity: small context movements produced large route changes. Version 1.1.2 therefore tests whether a bounded residual and a context-only continuity hinge can preserve route structure without recreating the sharp local tails.

The central question is deliberately narrow:

> Given the same frozen context, the same fixed host functions, and the same common functional cost, does a linear-first bounded residual router improve held-out functional route geometry while remaining as smooth as the linear router and as operationally safe as the MLP router?

2.2 Frozen causal chain

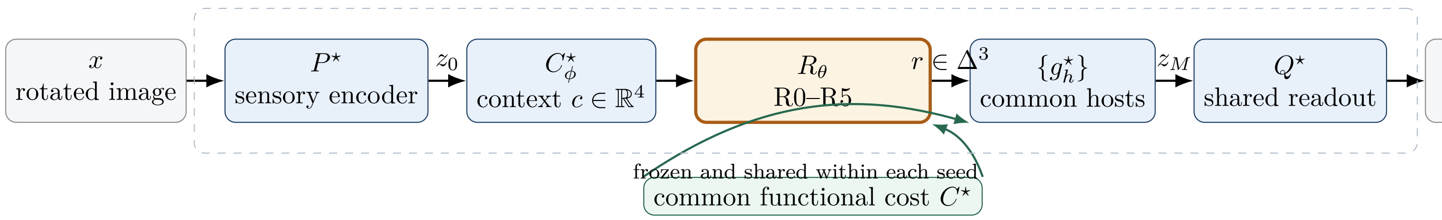

Figure 1: v1.1.2 bridge isolation. Only router parameters differ during the treatment comparison.

2.3 Six router treatments

Table 2: Router treatments and trainable parameter counts.

Router	Parameters	Role
R0 MLP	4,740	Flexible operational reference.
R1 linear	20	Low-capacity smooth structural reference.
R2 prototype energy	24	Local prototype territories.
R3 hybrid energy	42	Directional plus prototype control from v1.1.1.
R4 smooth residual	141	Linear base plus bounded prototype residual.
R5 smooth residual + continuity	141	R4 with context-only simplex-continuity hinge.

3 Mathematical objects

3.1 Route simplex

For four hosts, the route satisfies

$$r = (r_1, r_2, r_3, r_4) \in \Delta^3, \qquad r_h \geq 0, \qquad \sum_{h=1}^{4} r_h = 1. \tag{1}$$

The nominal distance uses the square-root simplex embedding:

$$d_R(r_a, r_b) = \frac{\left\|\sqrt{r_a} - \sqrt{r_b}\right\|_2}{\sqrt{2}}. \tag{2}$$

This is a Hellinger-type distance rather than a Euclidean distance between unconstrained logits.

3.2 Common functional route distance

A route is interpreted through the frozen host ecosystem. Let $C^\star_{hk}$ denote the normalized output distance between hosts h and k. The functional route distance is an entropically regularized optimal-transport cost

$$d_F(r_a, r_b) = \mathrm{OT}_{C^\star,\varepsilon}(r_a, r_b). \tag{3}$$

Every treatment is evaluated with the same $C^\star$ within a seed. This is essential: otherwise a router could appear geometrically different simply because it trained a different host bank.

3.3 Smooth residual router

R4 and R5 start from a linear base

$$\ell^{\text{base}}(c) = Wc + b. \tag{4}$$

Prototype evidence is centered, normalized, and bounded:

$$\delta_h(c) = \tanh\left(\frac{s_h(c) - \bar{s}(c)}{\mathrm{Std}_k\, s_k(c) + \epsilon}\right), \qquad |\delta_h(c)| \le 1. \tag{5}$$

A gate $g(c) \in [0, 1]$ controls when the residual is active, and the residual amplitude is capped by $s_{\max}$:

$$r(c) = \mathrm{softmax}(Wc + b + s_{\max} g(c)\delta(c)) . \tag{6}$$

R5 adds a factor-label-free continuity hinge

$$\mathcal{L}_{\text{cont}} = \mathbb{E}\left[\max\left(0, d_R(r_a, r_b) - \kappa d_C(c_a, c_b) - m\right)^2\right] . \tag{7}$$

The hinge uses only inferred contexts and routes; it does not use the ground-truth rotation angle.

3.4 Covariance-aware context geometry

For each angle u, context vectors across source identities define a regularized covariance

$$\Sigma_u = (1-\lambda)\widehat{\mathrm{Cov}}(c \mid u) + \lambda \frac{\mathrm{tr}(\widehat{\Sigma}_u)}{d} I + \epsilon I. \tag{8}$$

The diagnostic compares centroid distance with three SPD distances:

$$d_{\text{LE}}(\Sigma_u, \Sigma_v) = \|\log \Sigma_u - \log \Sigma_v\|_F \,, \tag{9}$$

$$d_{\text{AIRM}}(\Sigma_u, \Sigma_v) = \left\|\log(\Sigma_u^{-1/2}\Sigma_v\Sigma_u^{-1/2})\right\|_F \,, \tag{10}$$

$$d^2_{\text{BW}}(\Sigma_u, \Sigma_v) = \mathrm{tr}\left(\Sigma_u + \Sigma_v - 2(\Sigma_u^{1/2}\Sigma_v\Sigma_u^{1/2})^{1/2}\right) . \tag{11}$$

The preregistered SPD candidate combines normalized centroid and Log-Euclidean distances. This axis is diagnostic; it does not claim that the network learned a global Riemannian manifold.

4 Protocol and statistics

The pilot uses model seeds 0–4. Every router receives 15,360 forward images and 80 optimizer steps. Source identities are split into disjoint partitions for common-cost construction, probe fitting, probe evaluation, tangent fitting, causal evaluation, and route-swap evaluation. Per-source geometry effects are bootstrapped within seed. Across seeds, mean effects use a Student-t interval with $n = 5$.

The primary structural contrast is R5 versus R1. R5 versus R0 is an operational and secondary geometric comparison. R5 versus R3 tests whether the repair removes sharpness without destroying the geometry that motivated the prototype family.

5 Primary router result

5.1 R5 does not robustly improve on R1

The primary held-out functional effects are

$$\Delta\rho_{R5-R1} = +0.010886, \qquad 95\%\ \mathrm{CI} = [-0.048685, 0.070456], \tag{12}$$
$$\Delta\mathrm{stress}_{R5-R1} = -0.017859, \qquad 95\%\ \mathrm{CI} = [-0.067115, 0.031396]. \tag{13}$$

The point estimates are favorable, but both intervals cross zero. The favorable seed fraction is 3/5 for correlation and 4/5 for stress. Under the preregistered rule, neither effect qualifies.

R4 also fails to improve on R1: its held-out functional correlation effect is -0.00435 with interval $[-0.07597, 0.06728]$, and stress effect is -0.00050 with interval $[-0.04803, 0.04703]$. The bounded residual alone therefore adds no measurable structural value over the linear route.

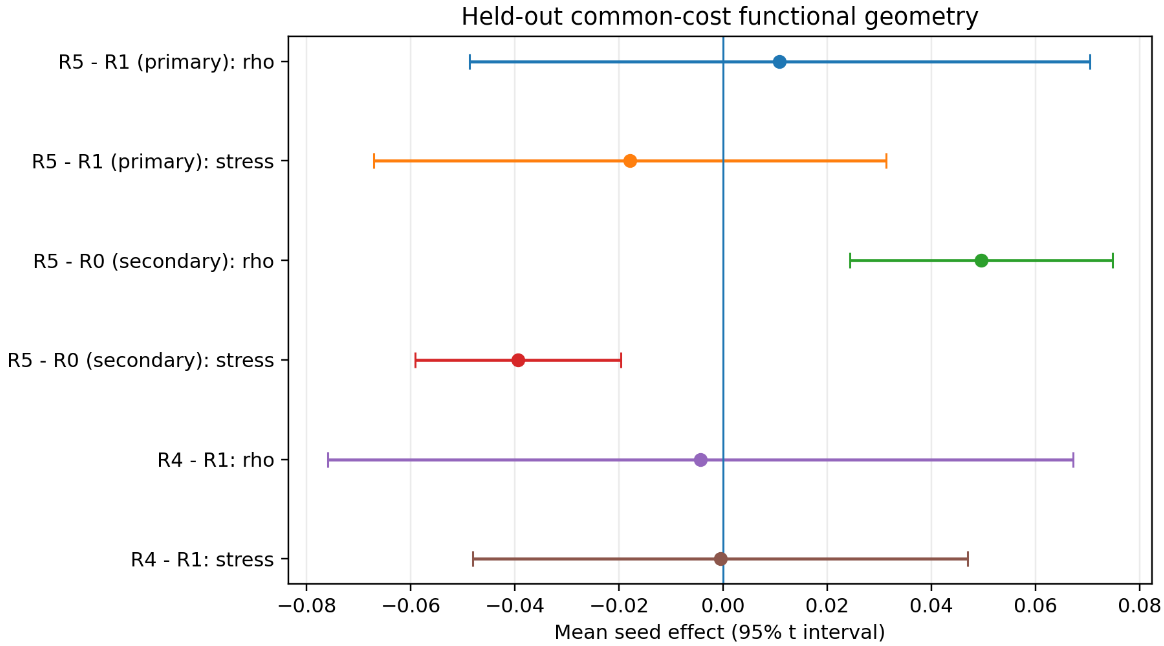

Figure 2: Held-out common-cost functional effects. For correlation, positive is favorable; for stress, negative is favorable. The R5-versus-R1 primary intervals cross zero. R5-versus-R0 is consistently favorable.

5.2 R5 significantly improves on R0

The secondary R5-versus-R0 comparison is unambiguously favorable:

$$\Delta\rho_{R5-R0} = +0.049574, \qquad 95\%\ \mathrm{CI} = [0.024319, 0.074828], \tag{14}$$
$$\Delta\mathrm{stress}_{R5-R0} = -0.039398, \qquad 95\%\ \mathrm{CI} = [-0.059165, -0.019631]. \tag{15}$$

All five seeds favor R5. This matters, but it does not rescue the primary claim. The correct interpretation is that the structured low-capacity family is more geometrically ordered than the flexible MLP, while the smooth residual does not improve robustly beyond the linear router.

Reviewer interpretation

The result is not "R5 does nothing." It is "R5 does not beat the simplest structured router." The linear bridge already captures most of the measurable context-to-route order. The residual and continuity machinery reduce sharpness but do not add robust held-out order beyond that baseline.

6 Continuity repair

The sharpness repair succeeds relative to R3. Mean held-out functional continuity q95 values are

Router	Mean q95
R5 residual + continuity	0.2693
R1 linear	0.2713
R4 residual	0.3108
R0 MLP	0.3436
R2 prototype	1.7180
R3 hybrid	1.9449

R5 reduces R3 q95 by 1.6755 on average with paired interval $[-1.7964, -1.5547]$. This is a clear mechanistic repair. Against R4, R5 reduces q95 by 0.0414 on average, but the interval $[-0.0903, 0.0074]$ still crosses zero.

Against the primary R1 reference, the mean difference is almost zero:

$$\Delta q95_{R5-R1} = -0.001924, \qquad 95\%\ \mathrm{CI} = [-0.170151, 0.166303]. \tag{16}$$

The upper interval exceeds the preregistered tolerance of 0.10, so the continuity non-degradation gate fails. Two seeds show R5 worse than R1, including one large positive deviation.

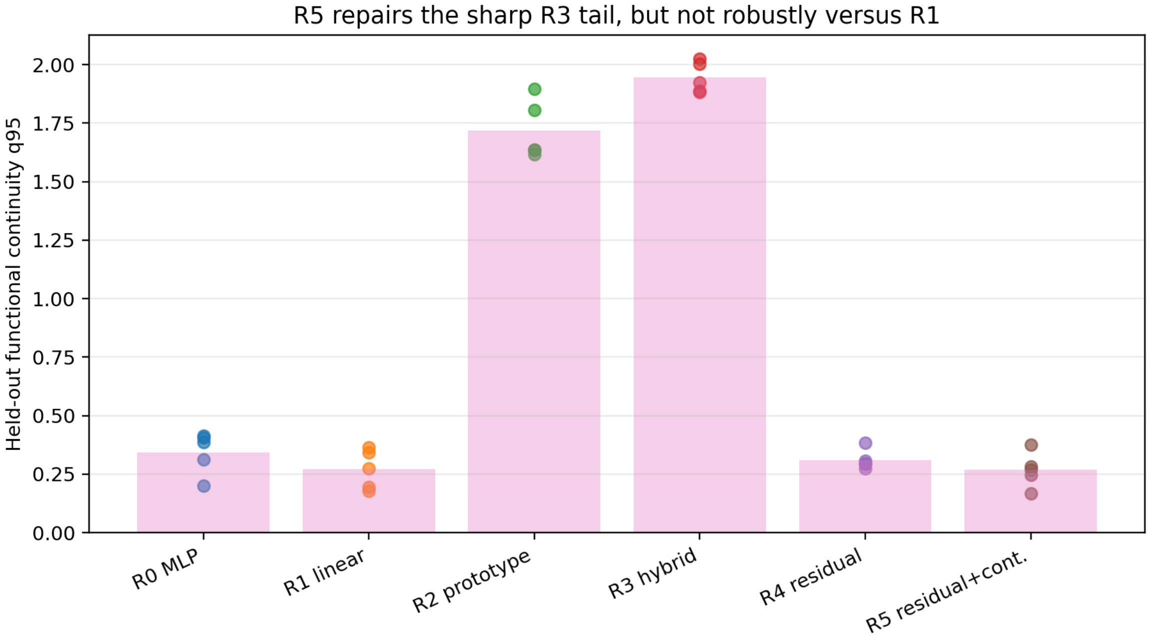

Figure 3: Held-out functional continuity q95. Dots are individual seeds; bars are means. R5 removes the R3/R2 sharpness but is not robustly non-degrading relative to R1.

7 Route swaps and causal interventions

7.1 Route-swap ordering passes

For R5, mean functional route distance is 0.05147 for near swaps and 0.11029 for far swaps, a ratio of 2.143. Far exceeds near in all five seeds. This supports a narrow causal-structural statement: the learned routes are not arbitrary labels, because swapping a route across a larger factor gap creates a larger functional displacement under the common host cost.

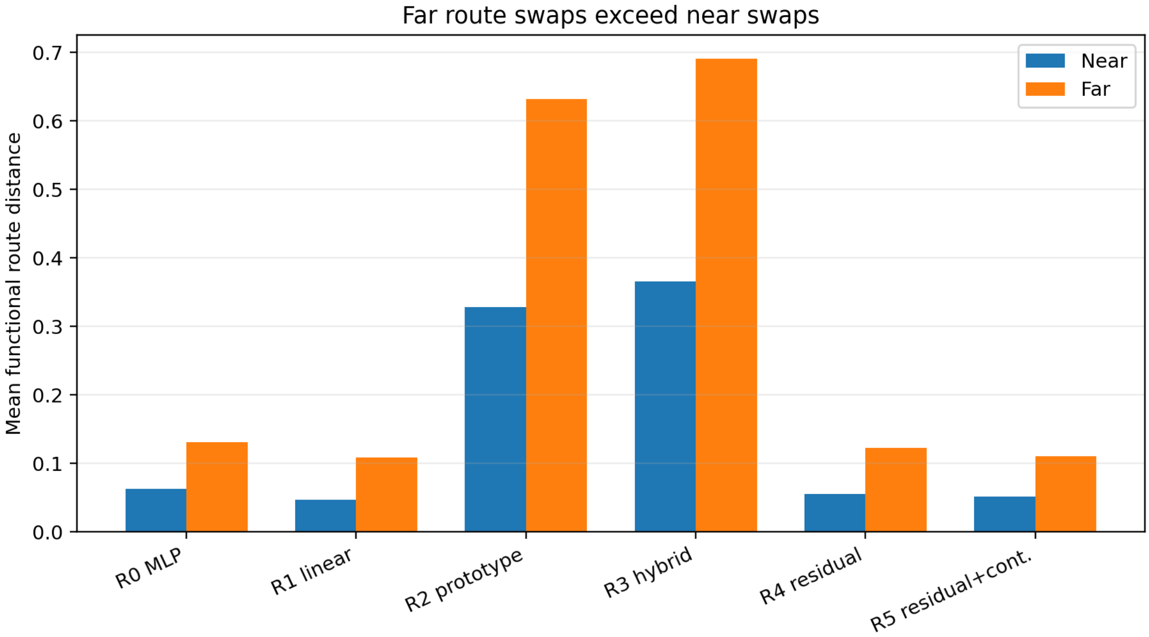

Figure 4: Mean functional distance for near and far route swaps. The ordering passes for R5, but swap order alone does not prove tangent-specific causal mediation.

7.2 Causal specificity fails

The causal specificity ratio compares route change under a fitted factor tangent with matched orthogonal controls. R5 mean functional CSR values by seed are 5.165, 0.430, 1.676, 1.620, and 0.010. The overall mean is 1.780, but the preregistered gate requires at least four of five seeds to have a lower bootstrap bound above one. Only seeds 0, 2, and 3 satisfy that condition. Seeds 1 and 4 show near-zero or sub-control specificity.

Prediction identity preservation is stable: the minimum per-seed preservation is 0.9811, so the identity gate passes. This combination means that interventions are usually small enough not to change the class, but their direction-specific functional effect is not stable across seeds.

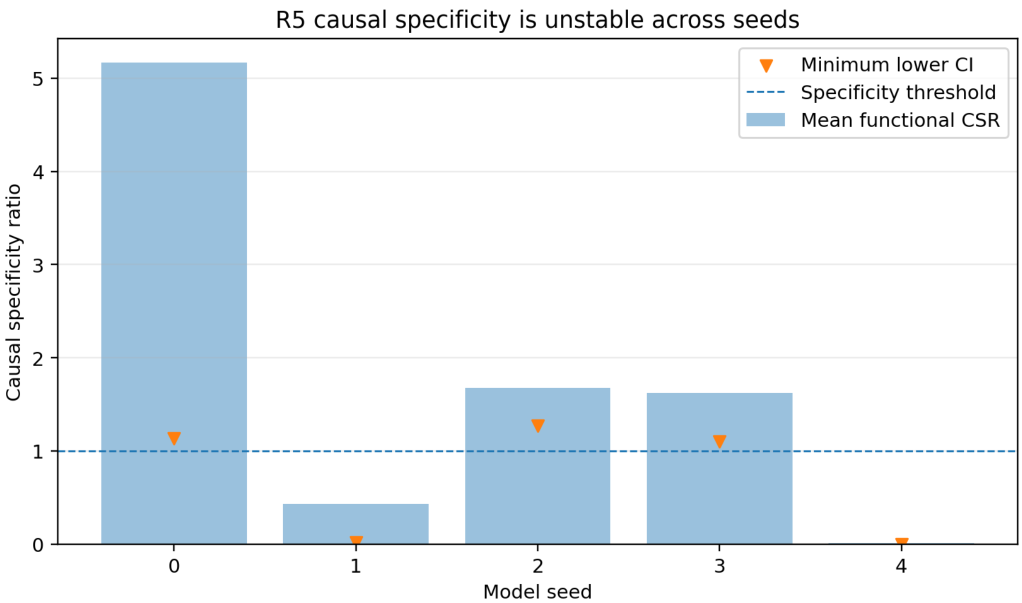

Figure 5: R5 causal specificity. Bars show mean CSR; triangles show the minimum lower bootstrap bound across intervention scales.

8 Operational metrics

R5 mean held-out accuracy is 0.56130 versus 0.57094 for R0, a loss of 0.00964 or 0.964 percentage points. The mean lies just inside the one-point equivalence margin, but equivalence is evaluated per seed: only two of five seeds remain within the margin. Final trained accuracy decreases by 0.00894 on average. Forgetting improves from 0.00200 for R0 to 0.000313 for R5, an improvement of 0.00169 or 0.169 percentage points.

The result therefore exposes a familiar tradeoff: the structured route is more stable and forgets less, but the shared frozen ecosystem loses classification accuracy. Since the preregistered operational gate requires both safety and structural evidence, the reduction in forgetting cannot compensate for the accuracy failures.

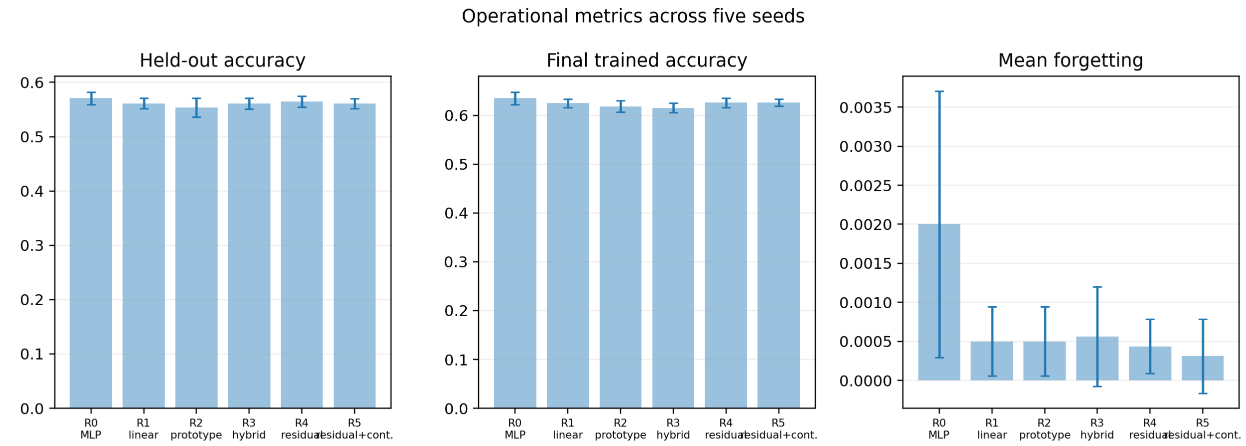

Figure 6: Five-seed operational metrics. Error bars are 95% seed intervals. R5 has low forgetting but lower accuracy than R0.

9 Covariance-aware context diagnostic

The SPD diagnostic is the clearest positive v1.1.2 result. Across seeds, centroid-only context geometry has mean $\rho = 0.5394$, stress 0.4898, and near/far AUC 0.8568. The combined centroid-plus-Log-Euclidean metric has mean $\rho = 0.6145$, stress 0.4456, and AUC 0.9060.

The preregistered stress contrast is

$$\Delta\text{stress}_{\text{combined}-\text{centroid}} = -0.044204, \qquad 95\%\ \text{CI} = [-0.063801, -0.024607], \tag{17}$$

with all five seeds favorable. Pure Log-Euclidean, AIRM, and Bures-Wasserstein distances have even lower mean stress, approximately 0.410, but the combined metric was the preregistered candidate because it retains both centroid displacement and covariance change.

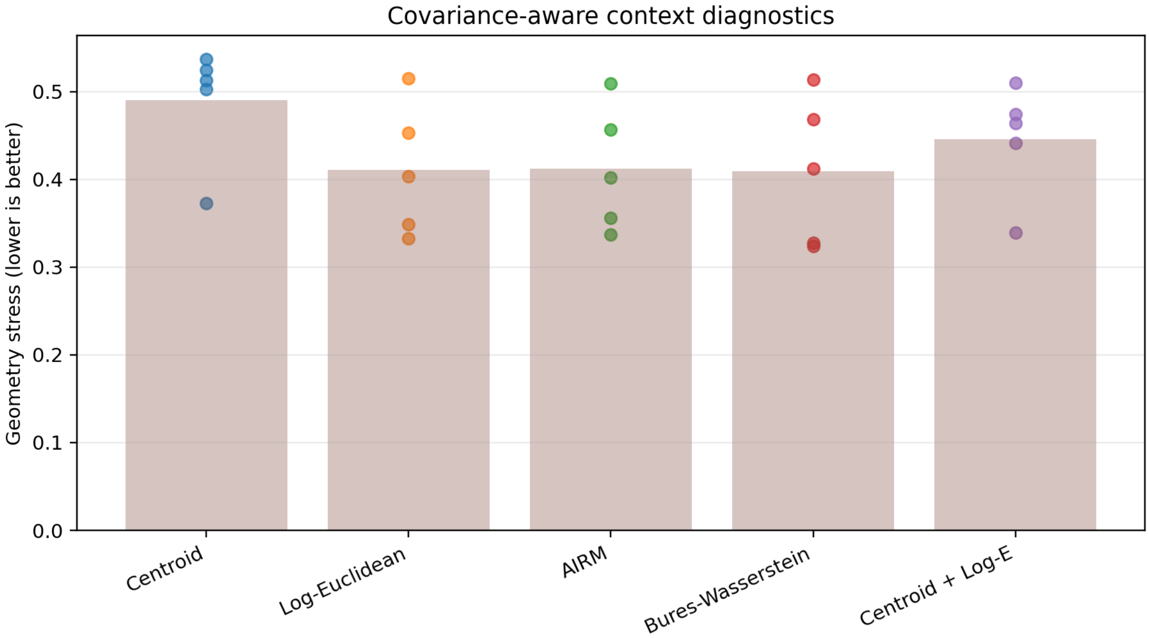

Figure 7: Context-geometry stress by metric and seed. Covariance-aware metrics generally reduce stress. The combined candidate improves over centroid-only geometry in all five seeds.

Claim boundary

The SPD result supports only the statement that context covariance contains stable factor information beyond the centroid. It does not show that the router uses this information, that the context space is globally Riemannian, or that an SPD-aware router would improve continual learning.

10 Simplex trajectories and host allocation

The route trajectories provide a visual explanation of the quantitative results. Prototype routers produce larger territorial motions. R5 lies closer to the compact linear/MLP region than R3, while retaining a more structured progression than R0 in several seeds.

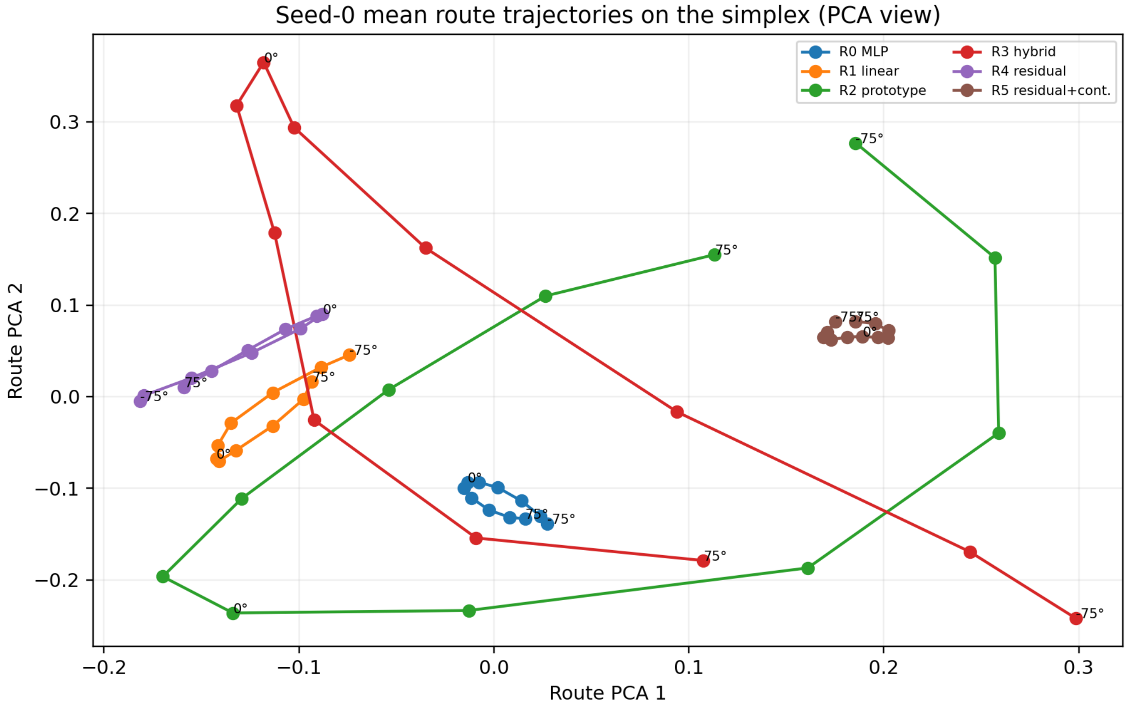

Figure 8: Seed-0 mean route vectors projected to two principal components. The data remain simplex-valued in four dimensions; PCA is used only for visualization.

Because hosts are frozen, these trajectories describe allocation territories, not learned host specialization. A router may consistently allocate factor regimes to different hosts even when those hosts were not causally specialized by the current stage. The report therefore keeps mature host specialization as not tested.

11 What v1.1.2 contributes

11.1 Positive contributions

1. It confirms that the v1.1.1 isolation protocol is reproducible with six router treatments.
2. It demonstrates that the sharp prototype continuity failure can be repaired by a bounded residual plus continuity hinge.
3. It shows that R5 and R1 are both significantly more ordered than the flexible MLP on common-cost held-out geometry.
4. It preserves route-swap ordering and prediction identity.
5. It establishes a stable covariance-aware context diagnostic across all five seeds.

11.2 Negative contributions

1. The residual adds no robust geometry beyond the linear router.
2. The continuity hinge is seed-unstable relative to R1.
3. Tangent-specific causal mediation remains unstable.
4. The fixed ecosystem loses accuracy relative to R0.
5. Better context geometry does not automatically become better functional routing.

12 Implications for the MMALS architecture

The two consecutive router studies now constrain the design space. v1.1.1 showed that strong prototypes are too sharp. v1.1.2 showed that smoothing those prototypes returns the system close to the linear baseline without producing a new robust gain. The immediate problem is therefore unlikely to be solved by adding another router term.

Three deeper hypotheses remain:

1. **Host-bank insufficiency.** The frozen common bank may not contain functionally distinct transformations aligned with the context factor.
2. **Readout insensitivity.** The shared classifier may not translate route differences into sufficiently useful output differences.
3. **Credit mismatch.** Classification and retention losses may not provide the router with causal allocation credit, even when context geometry is ordered.

These hypotheses correspond directly to earlier MMALS ideas about bidirectional information flow. The next stage should not immediately unfreeze everything. It should introduce controlled diagnostics of host usefulness, route-function transport, and selective host adaptation while preserving matched-compute and source-disjoint evaluation.

13 Recommended roadmap

13.1 v1.1.3: route-function memory transport

The positive SPD diagnostic should be used to characterize memory trajectories, not to create another untested router. Let μ_t denote the distribution of route-function states after learning stage t. A transport diagnostic can ask how much work moves old memory into new memory:

$$\mathcal{T}_c(\mu_t, \mu_{t+1}) = \inf_{\gamma \in \Pi(\mu_t, \mu_{t+1})} \int c(u, v)\, d\gamma(u, v). \tag{18}$$

The cost should separate route-only, latent-functional, behavioral, and hybrid components. This tests whether large transport predicts forgetting and whether path reconstruction is possible.

13.2 v1.1.4: selective causal host adaptation

The frozen-bank experiment cannot establish mature specialization. A later stage should selectively unfreeze hosts under explicit contribution and recovery controls. The core test is whether the host with the largest route allocation also has the largest causal ablation effect and whether targeted adaptation improves that alignment without destroying retention.

13.3 G2.1 and TPUT

Explicit host-to-router feedback and goal-conditioned control remain future layers. They should be introduced only after the program can measure host confidence, novelty, capacity, and contribution independently. Otherwise, feedback would add another controller without resolving the current functional bottleneck.

14 Claims and non-claims

Statement	Status	Evidence boundary
Common context and host bank are identical across routers	Supported	Exact preservation audit, five seeds.

Statement	Status	Evidence boundary
Router compute is matched	Supported	Equal forward images and optimizer steps.
R5 improves held-out functional geometry over R0	Supported, secondary	Both seed intervals exclude zero.
R5 improves held-out functional geometry over R1	Not supported	Primary intervals cross zero.
R5 removes R3 continuity sharpness	Supported, mechanistic	Large negative paired q95 difference.
R5 is non-degrading in continuity relative to R1	Not supported	Upper paired interval exceeds tolerance.
Far route swaps exceed near route swaps	Supported	All five R5 seeds.
R5 has stable tangent-specific causal mediation	Not supported	Only three of five seeds meet lower-bound criterion.
R5 preserves prediction identity under small interventions	Supported	Minimum preservation above 0.98.
R5 is operationally non-degrading relative to R0	Not supported	Per-seed accuracy margin fails.
Covariance adds context-geometry information	Supported, diagnostic	Combined stress improves in all five seeds.
MMALS learned a Riemannian manifold	Not claimed	SPD matrices are post-hoc diagnostics.
Hosts are mature specialists	Not tested	Hosts are frozen.
Memory transport reduces forgetting	Not tested	Deferred to v1.1.3.
Operational superiority	Not claimed	Not preregistered.
Quantum advantage	Not claimed	Outside the experiment.

15 Limitations

The pilot uses RotatedMNIST, five model seeds, and a four-dimensional inferred context. The host bank and classifier are deliberately frozen, which improves causal isolation but can reduce operational performance. The factor is one-dimensional and smooth; results may not transfer to class-incremental, domain-incremental, multimodal, or nonstationary environments. Across-seed intervals with $n = 5$ are necessarily wide. The SPD diagnostic uses factor-grouped covariances and therefore remains a post-hoc analysis rather than a task-free control signal. Finally, the positive R5-versus-R0 geometry result should not be promoted to a global MMALS claim because the preregistered structural reference is R1.

16 Conclusion

Geometry-MMALS G1 v1.1.2 completes a clean five-seed test of smooth simplex-residual routing under a frozen functional ecosystem. The router repair succeeds in the narrow sense that it eliminates the severe local sensitivity of the prototype-hybrid R3 while preserving non-arbitrary route structure. It also significantly outperforms the flexible MLP on held-out common-cost route geometry. However, it does not robustly improve on the linear router, does not achieve stable causal specificity, and does not satisfy operational non-degradation. R5 therefore does not qualify.

The positive covariance diagnostic changes the next question. The project now has evidence that context distributions carry factor information beyond their centroids, but not that the current router or host bank can exploit it. The scientifically disciplined next step is route-function memory transport followed by selective causal host adaptation, not another increasingly elaborate router.

A Exact primary effects

Table 4: Held-out functional contrasts over five seeds.

Contrast	Metric	Mean	95% low	95% high
R5–R1	ρ	0.010886	-0.048685	0.070456
R5–R1	stress	-0.017859	-0.067115	0.031396
R5–R0	ρ	0.049574	0.024319	0.074828
R5–R0	stress	-0.039398	-0.059165	-0.019631
R4–R1	ρ	-0.004346	-0.075974	0.067282
R4–R1	stress	-0.000498	-0.048029	0.047034
R5–R3	ρ	0.027786	-0.028377	0.083948
R5–R3	stress	0.002200	-0.028963	0.033362

B Reproducibility record

- Package version: `1.1.2`.
- Executed build revision: `smooth-residual-spd-diagnostics-c0-r1`.
- Git SHA recorded by the run: `c5106d1c7f7e99c8b6de9b1880d21c1dbe00ad6e`.
- Model seeds: 0, 1, 2, 3, 4.
- Sensory accuracy: 0.942.
- Continuity loss uses no external factor labels.
- The common functional metric and frozen host bank are shared across treatments.
- Corrected manifests are metadata-only; numerical results are unchanged.